

KIRTHANA P RAO

Email: work.kirthana@gmail.com | Place: Bangalore, India

+91 94970-24531
[LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Motivated and detail-oriented ECE graduate, seeking a challenging role in Systems Design/Verification. Dedicated to continuous learning and eager to research and contribute, with keen interest in Embedded Systems and at the intersection of AI and VLSI, with interests in Cybersecurity, Healthcare Technology and Sustainable Manufacturing.

TECHNICAL SKILLS

- Languages:** C, C++, Python, Verilog, SystemVerilog, TL-Verilog HDL
- Hardware Tools:** Cadence Virtuoso, Xilinx Vivado, KiCAD, MATLAB Simulink, OpenLANE, OpenSTA, LTSpice
- Platforms & Frameworks:** RISC-V ISA, Linux OS, AXI4-Stream/AXI-4 Lite Protocol, TensorFlow 2.0, ONNX, FINN Compiler
- Core Concepts:** Digital Design, Machine Learning, Agile Methodologies, Data Structures & OOP, Timing Analysis, PCB Design

EDUCATION

Indian Institute of Information Technology, Design & Manufacturing, (IIITDM) Kurnool Nov 2022 – Jul 2026
B.Tech Electronics & Communication Engineering with Specialization in Design and Manufacturing · CPI: 7.82/10.00

EXPERIENCE

- Pre-Doctoral Fellow, Indian Institute of Science (IISc), Bangalore** July 2026 – Present
Affiliated with the Walmart Centre of Technical Excellence, IISc. Working on optimizing performance within AMD Versal devices such as V80 using Network-on-Chip Interconnects for ML Algorithms, at the FIST (Future Computing Systems) Lab, Dept. of Computer Science and Automation, in collaboration with AMD R&D.
- Research Intern, Indian Institute of Science (IISc), Bangalore** Jan 2026 – Present
Working on designing an efficient architecture for Hardened Network-on-Chip based ML Accelerator, on the AI Model Optimization end and Acceleration Design, using ONNX graph, at the FIST (Future Computing Systems) Lab, Dept. of Computer Science and Automation
- Summer Intern on ‘Digital Systems Prototyping using FPGAs’, NIELIT Calicut, Kozhikode** May 2025 – Jun 2025
Completed an intensive training in RTL Design and FPGA-based logic synthesis, and performed RTL coding, testbench development, and synthesis workflows for FPGA implementations, to create and verify the designs meeting performance and functionality.

PROJECTS

- AXI4-Lite Protocol Based Packet Sorter** Jun 2025
An AXI4-Lite Protocol transaction is programmed in Slave Peripheral and sorts 32-bit data packets into valid and invalid data using two simple FIFOs as storage.
 - Key Skills:** AXI4-Lite Protocol · RTL Design · Verilog
- Design of Compressor Based Dadda Multiplier** Mar 2025 – Apr 2025
Designed and implemented 4-2 and 3-2 compressors for large-scale multiplier applications- an 8x8 bit Dadda Multiplier, in Xilinx Vivado 2024.2v. Resulted in the power reduction by about 8%, and better utilization of area.
 - Key Skills:** RTL Design · Verilog · Digital Logic Design
- Design and Simulation of RISC-V ALU and a Pipelined RISC-V Processor** May 2025
Created a robust implementation of a multi-stage pipelined RISC-V processor emulator featuring comprehensive hazard detection and resolution mechanisms. Designed and debugged a single-cycle RISC-V ISA implementation using GCC compilation and Spike simulator validation, covering instruction decoding, register file management, ALU operations, and control flow hazard mitigation.
 - Key Skills:** TL-Verilog · C · RISC-V · Verilog

CERTIFICATIONS

- Digital Design with Verilog, **NPTEL** and C-Based VLSI Design, **NPTEL** by **IIT Guwahati**
- VLSI Design Flow: RTL to GDS, **NPTEL** by **IIIT Delhi**
- Machine Learning Specialization by **Stanford University & DeepLearning.AI** on **Coursera**
- Neural Networks and Deep Learning and Convolutional Neural Networks by **DeepLearning.AI** on **Coursera**
- Minor Degree in Internet of Things (IoT), **IIITDM Kurnool**

ACHIEVEMENTS

- Ranked among the top 0.1% of Computer Science students in the Secondary School CBSE Board Examination.
- Second runner-up as a team in RoboRhythm competition - in building a Line Following Robot in SOLASTA Techno-Cultural Fest 2K24
- ISWDP Samsung Student Fellowship Recipient
- SPARKS Predoctoral Research Fellowship, Indian Institute of Science, 2026

LANGUAGES KNOWN

English, Hindi, Kannada, Malayalam, Konkani, Telugu

EXTRA CURRICULAR

Technical Coordinator at *ElectroniX Club*, IIITDM Kurnool · *Roohaniyat Music Club*, IIITDM Kurnool
Carnatic Vocal Music – Passed with *First Division* in Vocal Music by *Bangiya Sangeet Parishad* 2018
Orator and Emcee